

AF-F7 mini

User Guide



AF-F7 mini Overview

F7 Mini Flight Controller

AF-F7 mini Specifications

• Basic Specifications

MCU	STM32F765, ARM Cortex-M7, 216 MHz
RAM / Flash	512 KB / 2 MB
IMU	ICM-20689 (Accel/Gyro)
Secondary IMU	ICM-20602, BMI055
Magnetometer	IST8310
Barometer	MS5611
Operating Voltage	4.75 – 5.5 V (Rated 5 V)
USB Input	4.75 – 5.25 V
Servo Rail	Max. 36 V (No Internal Regulator)
PWM Output	8 FMU channels + 3 auxiliary channels in current ArduPilot configuration

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RC Input	S.Bus, PPM, DSM/DSM2/DSMX, CPM
RSSI Input	Analog / PWM
UART	6 hardware serial interfaces configured; verify connector pinout
Serial Mapping	SERIAL1 = USART2 · SERIAL2 = USART3 · SERIAL3 = USART1 · SERIAL4 = UART4 · SERIAL5 = USART6 · SERIAL6 = UART7 (USB = SERIAL0)
I ² C	4 buses configured, including internal sensor buses
CAN	2 Port
ADC	VBat/Current + Aux Analog Input 2Ch

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Size	64.2 × 42.3 × 14.6 mm
Weight	39.7 g
Operating Temp	-20 ~ +85 °C
Supported F/W	novaX ArduPilot (Copter and Plane releases)

AF-F7 mini Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 4.75 – 5.5 V (Rated 5 V).
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-F7 mini Firmware

• Available Firmware

Type	Version	Date
ArduPilot Copter (.apj package)	1.3.0	2026-08-31
ArduPilot Copter - Bootloader + App (merged HEX / DFU / SWD)	1.3.0	2026-08-31
ArduPilot Plane (.apj package)	1.3.0	2026-08-31
ArduPilot Plane - Bootloader + App (merged HEX / DFU / SWD)	1.3.0	2026-08-31

Download files from the product page (QR).

AF-F7 mini Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.



<Detailed Manual>

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